

Chapter 3 Lab: Mathematical Operations with Arrays

Instructions:

- Write a script file to do the exercises below.
- Make sure to write comments in your code and separate output display for clarity.
- Submit your script file in the following format: FirstName_LastInitial_ch3lab.m
Ex: Ryan Cao would submit his lab as: **ryan_c_ch3lab.m**

1. Define the vectors:

$$\mathbf{u} = 5\mathbf{i} - 2\mathbf{j} + 4\mathbf{k} \quad \mathbf{v} = -2\mathbf{i} + 7\mathbf{j} + 3\mathbf{k} \quad \mathbf{w} = 8\mathbf{i} + 1\mathbf{j} - 3\mathbf{k}$$

Use the vectors to verify the identity:

$$(\mathbf{u} + \mathbf{v}) \cdot [(\mathbf{v} + \mathbf{w}) \times (\mathbf{w} + \mathbf{u})] = 2\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$$

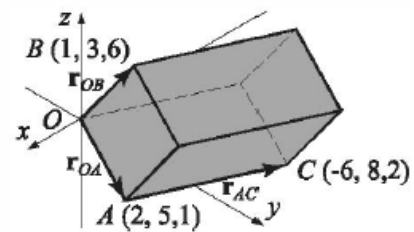
Use MATLAB's built-in function cross and dot to calculate the value of the left and right sides of the identity.

2.

The volume of the parallelepiped shown can be calculated by $\mathbf{r}_{OB} \cdot (\mathbf{r}_{OA} \times \mathbf{r}_{AC})$. Use the following steps in a script file to calculate the area.

Define the vectors $\mathbf{r}_{OA}$, $\mathbf{r}_{AC}$, and $\mathbf{r}_{OB}$ from knowing position of points A , B , and C .

Determine the volume by using MATLAB's built-in functions dot and cross.



3.

Show that $\lim_{x \rightarrow 0} \frac{\cos(2x) - 1}{\cos x - 1} = 4$.

Do this by first creating a vector x that has the elements 1.0, 0.5, 0.1, 0.01, 0.001, and 0.0001. Then, create a new vector y in which each element is determined from the elements of x by $\frac{\cos(2x) - 1}{\cos x - 1}$. Compare the elements of y with the value 4 (use format long to display the numbers).

4. Create the following matrices:

$$A = \begin{bmatrix} 1 & -3 & 5 \\ 2 & 2 & 4 \\ -2 & 0 & 6 \end{bmatrix} \quad B = \begin{bmatrix} 0 & -2 & 1 \\ 5 & 1 & -6 \\ 2 & 7 & -1 \end{bmatrix} \quad C = \begin{bmatrix} -3 & 4 & -1 \\ 0 & 8 & 2 \\ -3 & 5 & 3 \end{bmatrix}$$

- a) Does $A * B = B * A$?
- b) Does $A * (B * C) = (A * B) * C$?
- c) Does $(A * B)^T = A^T * B^T$ Note: A^T means A transpose
- d) Does $(A + B)^T = A^T + B^T$?

5.

Solve the following system of three linear equations:

$$-4x + 3y + z = -18.2$$

$$5x + 6y - 2z = -48.8$$

$$2x - 5y + 4.5z = 92.5$$

Outputs:

Problem 1:

LHS =
-776

RHS =
-776

Problem 2:

Volume =
248

Problem 3:

x =

Columns 1 through 3

1.0000000000000000 0.5000000000000000 0.1000000000000000

Columns 4 through 6

0.0100000000000000 0.0010000000000000 0.0001000000000000

each_result =

Columns 1 through 3

3.080604611736280 3.755165123780746 3.990008330556008

Columns 4 through 6

3.999900000832619 3.999999000133061 4.000000000000000

Problem 4:

Part (a)

$p_1 =$

-5 30 14
18 26 -14
12 46 -8

$p_2 =$

-6 -4 -2
19 -13 -7
18 8 32

No

Part (b)

$v_1 =$

-27 290 107
-12 210 -8
-12 376 56

$v_2 =$

-27 290 107
-12 210 -8
-12 376 56

Yes

Part (c)

$t_1 =$

-5 18 12
30 26 46
14 -14 -8

$t_2 =$

-6 19 18
-4 -13 8
-2 -7 32

No

Part (d)

$s_1 =$

1 7 0
-5 3 7
6 -2 5

s2 =
1 7 0
-5 3 7
6 -2 5

Yes

Problem 5:

result =

2.8000

-6.4000

12.2000